Reward Recovery in Sequential Decision Models

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1 Setup

Consider a stationary infinite-horizon MDP with finite state space $\mathcal{S}$, finite action space $\mathcal{A}$, discount factor $\beta \in (0,1)$, transition kernel $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, and unknown reward $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$. Additive T1EV shocks yield the soft Bellman recursion.

$$Q^*(s,a) = r(s,a) + \beta \sum_{s'} P(s' \mid s,a)\, V^*(s'), \tag{1}$$

$$V^*(s) = \log \sum_{a'} \exp Q^*(s,a'), \tag{2}$$

$$\pi^*(a \mid s) = \exp\bigl(Q^*(s,a) - V^*(s)\bigr). \tag{3}$$

The advantage $A^*(s,a) = Q^*(s,a) - V^*(s) = \log \pi^*(a \mid s)$ is identified from CCPs. The reward r is not identified in levels: any $\tilde{r}(s,a) = r(s,a) + \lambda(s)$ generates identical CCPs (Magnac and Thesmar, 2002). The set of all such $\tilde{r}$ is the identified set (Kalouptsidi et al., 2021).

2 Four Approaches

2.1 Reduced-form Q

The reduced-form approach estimates $Q_\omega(s,a)$ by logit MLE.

$$\hat{\omega} = \arg\max_{\omega} \sum_{i,t} \log \frac{\exp Q_\omega(s_{it}, a_{it})}{\sum_{a'} \exp Q_\omega(s_{it}, a')}. \tag{4}$$

At the population optimum, $Q_{\hat{\omega}}(s,a) = Q^*(s,a) + c(s)$ for some action-independent c, so the backed-out "reward" is $A^*(s,a)$, not $r(s,a)$. No Bellman equation is imposed and no transition data are used.

2.2 AIRL (Fu et al., 2018)

The AIRL discriminator decomposes as follows.

$$f_\theta(s,a,s') = g_\theta(s,a) + \beta\, h_\phi(s') - h_\phi(s). \tag{5}$$

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Here g_θ approximates the reward and $\beta h(s') - h(s)$ is potential-based shaping (Ng et al., 1999). At the GAN optimum the discriminator score equals the advantage, $f^* = A^*$ (Fu et al., 2018, Appendix A.4), which implies the following expression for the recovered reward network.

$$g^*(s,a) = r(s,a) - \delta(s) + \beta \sum_{s'} P(s' \mid s,a)\, \delta(s'), \tag{6}$$

where $\delta(s) \equiv V^*(s) - h^*(s)$ is the shaping residual. Unless $\delta \equiv 0$, the recovered g^* is a potential-based perturbation of r that generates the same policy in the original MDP but can produce different behavior under changed dynamics.

Fu et al. address this through a state-only restriction: if g_θ depends only on s (not a) and the dynamics satisfy a decomposability condition, then $g^*(s) = r(s) + \text{const}$, recovering a disentangled reward that transfers across dynamics. This is what enables their sim-to-real experiments. However, the state-only restriction requires $r(s,a) = r(s)$, which fails whenever rewards depend on actions (e.g., paying vs. waiting).

2.2.1 LSW anchor identification (Lee et al., 2026)

Lee et al. (2026) (henceforth LSW) take a different approach. They allow action-dependent rewards and impose two anchors instead. First, the exit payoff is normalized to zero, so $g(s, \text{exit}) = 0$ for all s. Second, the shaping function is pinned at the absorbing state, so $h(s_{\text{abs}}) = 0$, where exit deterministically transitions to s_{abs}.

Lemma 1 (Non-identifiability)**.** *If (g,h) and $(\bar{g},\bar{h})$ produce the same f for all (s,a,s'), then $\bar{h} = h + \Phi$ and $\bar{g}(s,a) = g(s,a) + \Phi(s) - \beta\Phi(s')$ for some $\Phi : \mathcal{S} \to \mathbb{R}$.*

Proof. Rearrange $g + \beta h(s') - h(s) = \bar{g} + \beta\bar{h}(s') - \bar{h}(s)$ and define $\Phi = \bar{h} - h$. □ □

Theorem 2 (Anchor identification)**.** *Under the two anchors, if (g,h) and $(\bar{g},\bar{h})$ realize the same f, then $g = \bar{g}$ and $h = \bar{h}$.*

Proof. By Lemma 1, $\bar{g}(s, \text{exit}) = 0 + \Phi(s) - \beta\Phi(s_{\text{abs}})$. The exit anchor forces $\Phi(s) = \beta\Phi(s_{\text{abs}})$ for all s. The absorbing anchor forces $\Phi(s_{\text{abs}}) = 0$. Hence $\Phi \equiv 0$. □ □

Theorem 3 (Recovery of true reward)**.** *If the true pair (r^*, V^*) satisfies $r^*(s, \text{exit}) = 0$ and $V^*(s_{\text{abs}}) = 0$, then any AIRL pair (g,h) realizing A^* under the same anchors satisfies $g = r^*$ and $h = V^*$.*

Proof. Define $\Phi = h - V^*$. The exit anchor gives $\Phi(s) = \beta\Phi(s_{\text{abs}})$. The absorbing anchor applied to both h and V^* gives $\Phi(s_{\text{abs}}) = 0$. □ □

The Kang et al. (2025) ERM framework uses the same anchor principle but solves a single non-adversarial optimization instead of a GAN.

The LSW model estimates segment-specific reward and policy functions for K latent consumer types. Each user belongs to segment $z \in \{1, \ldots, K\}$, and the objective is to recover the segment-specific reward $r^*(s,a,z)$ and policy $\pi^*(a \mid s,z)$ that rationalize observed trajectories under the entropy-regularized Bellman equation. Estimation proceeds via an EM algorithm: the E-step computes posterior segment membership from trajectory likelihoods, and the M-step updates each segment's reward via AIRL with the anchor constraints applied per segment.

2.3 IQ-Learn (Garg et al., 2021)

The inverse Bellman operator $r_Q(s,a) \equiv Q(s,a) - \beta \sum_{s'} P(s' \mid s,a)\, V_Q(s')$ expresses reward as a function of Q alone. IQ-Learn estimates a single Q-function by solving the following optimization problem.

$$\max_Q \; \mathbb{E}_{(s,a)\sim\rho^*}\big[\phi\big(r_Q(s,a)\big)\big] - (1-\beta)\,\mathbb{E}_{s_0\sim\mu_0}\big[V_Q(s_0)\big], \tag{7}$$

The function $\phi(x) = x - x^2/2$ corresponds to χ^2 divergence. At the optimum, r_Q separates immediate reward from continuation values via the Bellman constraint. IQ-Learn requires the transition kernel $P(s' \mid s,a)$ to compute the inverse Bellman operator and uses it in the objective through $V_Q(s')$. It does not require adversarial training.

2.4 GLADIUS (Kang et al., 2025)

GLADIUS parameterizes both $Q_\theta(s,a)$ and $\hat{E}_\theta[V(s') \mid s,a]$ with neural networks, trained jointly on observed (s,a,s') transitions via the following loss.

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_i \log \pi_\theta(a_i \mid s_i) + \lambda \frac{1}{N}\sum_i \big(\hat{E}_\theta[V(s_i') \mid s_i, a_i] - V_\theta(s_i')\big)^2, \tag{8}$$

where $V_\theta(s') = \log \sum_{a'} \exp Q_\theta(s',a')$. After training, reward is extracted via the same inverse Bellman operator as IQ-Learn. Unlike IQ-Learn, GLADIUS does not require the transition matrix P at estimation time. It learns $\hat{E}[V(s') \mid s,a]$ from empirical next-state observations rather than computing it from known transitions. The MLP architecture shares parameters across states, providing implicit regularization that tabular Q parameterizations lack.

2.5 NFXP (Rust, 1987)

The nested fixed point (NFXP) estimator parameterizes the flow utility $u_\theta(s,a)$, solves the Bellman equation exactly under known transitions $P(s' \mid s,a)$, and maximizes the log-likelihood of observed actions via the implied softmax probabilities. We use a policy/value iteration hybrid to solve the inner fixed point and a BHHH/BFGS outer loop. Under correct specification, NFXP consistently recovers r^* and yields valid Type B counterfactuals; its trade-off is computational cost.

3 Counterfactual Taxonomy

Definition 4 (Type A: state extrapolation)**.** *The transition kernel P and reward r are unchanged. Only the realized state values shift. The counterfactual CCPs require only evaluating the estimated Q at new state inputs, with no Bellman re-solve.*

Definition 5 (Type B: environment change)**.** *The reward r is unchanged, but the transition kernel shifts to $\tilde{P}$ or the action set changes. The counterfactual Q must satisfy a new Bellman equation $\tilde{Q} = r + \beta\tilde{P}\tilde{V}$, which requires the structural reward r separated from the old continuation values.*

Proposition 6 (Type A equivalence)**.** *Under Type A, all methods considered yield identical counterfactual CCPs.*

Proof. Each method recovers scores whose within-state action differences match Q^*, which can be written as $Q^*(s,a) + c(s)$ for some action-independent c. For any shifted state $\tilde{s}$, the softmax $\pi(a \mid \tilde{s}) \propto \exp(Q^*(\tilde{s},a) + c(\tilde{s}))$ cancels $c(\tilde{s})$. □ □

Proposition 7 (Type B divergence)**.** *Under Type B, reduced-form and AIRL-without-anchors produce invalid predictions. AIRL with anchors, IQ-Learn, and GLADIUS produce valid predictions when the structural reward is correctly recovered.*

The reduced-form estimate $Q^* + c(s)$ bundles r with βPV^*, so under $\tilde{P}$ it cannot disentangle r. AIRL without anchors recovers $g^* = r + \text{shaping}$, which is observationally equivalent in the original MDP but produces incorrect behavior under $\tilde{P}$ unless $\delta \equiv 0$. AIRL with anchors recovers $g^* = r^*$ exactly (Theorem 3). IQ-Learn and GLADIUS separate r via the inverse Bellman operator. The LSW anchors provide what Kalouptsidi et al. (2021) call the "additional restrictions" needed to identify counterfactuals under transition changes.

4 Application to LSW (2026)

LSW study sequential content consumption with three actions (buy, wait, exit) using AIRL under the two anchors and deterministic transitions. Their wait-time validation (Section 6.3) is a Type A counterfactual, so every method that recovers the advantage function would pass it identically. Their WFF removal and segment-specific delay policies (Sections 7.1–7.2) are Type B because they change the action-set structure and the MDP that each segment faces. Their content-based paywall (Section 7.3) is also Type B because high-content episodes become paid-only, altering the transition structure episode-by-episode.

5 Simulation Study

Data sources. Analyses A–C and E–G report population-level results computed from the solved MDP or closed-form shaping (no estimation noise). The JSON artifact for these is `experiments/identification/r` which also drives the repository figures. Analysis D reports finite-sample outcomes using the estimator implementations in `econirl`.

We construct a tabular MDP that mimics the LSW book-reading environment. The state encodes episode position, content quality, and wait-time level, giving $15 \times 3 \times 4 = 180$ regular states plus one absorbing state. The three actions are buy, wait, and exit, with deterministic transitions and $\beta = 0.95$. The true reward is $r^*(s, \text{buy}) = u(c) - 2.0$, $r^*(s, \text{wait}) = u(c) - 0.08w$, and $r^*(s, \text{exit}) = 0$, where $u \in \{0.5, 1.5, 3.0\}$. Analyses A through C and E through F are evaluated at the population level with no estimation noise. Analysis D uses finite-sample estimation with the actual IQ-Learn and GLADIUS implementations from the `econirl` package.

5.1 A. Reward recovery

Methods with anchors or Bellman consistency pin the value function offset and recover r^* exactly. The reduced-form logit converges to A^*, which bundles reward with continuation values. AIRL without anchors recovers a shaped perturbation with $\delta = 0.5V^*$. All five methods produce Type A CCP errors below 10^{-6}, confirming Proposition 6.

Method	MSE vs. r^*	Corr. with r^*
Oracle / AIRL+anchors / IQ-Learn / GLADIUS	0.000	1.000
AIRL-no-anchors	10.005	0.440
Reduced-form	40.020	0.138

Table 1: Population-level reward recovery.

5.2 B. Type B error vs. transition shift

The counterfactual changes the transition kernel so that buying advances k episodes instead of one. The Bellman equation is re-solved under each method's recovered reward. The reduced-form approach cannot separate reward from continuation values, so it re-solves with A^* as a proxy for reward, which embeds the original transition structure. AIRL without anchors re-solves with the shaped reward g^*, which embeds the shaping residual.

k	Oracle	AIRL+a	IQ/GLADIUS	AIRL-no-a	RF
1	0.000	0.000	0.000	0.000	0.000
2	0.000	0.000	0.000	0.025	0.082
3	0.000	0.000	0.000	0.017	0.094
5	0.000	0.000	0.000	0.015	0.123
7	0.000	0.000	0.000	0.016	0.129
10	0.000	0.000	0.000	0.014	0.139

Table 2: Type B CCP error vs. oracle. All five methods are shown. Structural methods with correct reward recovery are exact at every k.

5.3 C. Shaping magnitude sweep

We construct shaped rewards $\tilde{r} = r^* + \alpha V^*(s) - \alpha\beta V^*(s')$ for $\alpha \in [0, 1]$ and evaluate the Type B counterfactual with $k = 3$. At $\alpha = 0$ the reward is exact and at $\alpha = 1$ it is maximally shaped.

α	Type B CF error	Corr. with r^*
0.00	0.000	1.000
0.10	0.004	0.908
0.30	0.011	0.610
0.50	0.017	0.440
1.00	0.027	0.270

Table 3: Counterfactual error is monotone in shaping magnitude. Without anchors, which α the optimizer finds is indeterminate.

5.4 D. Finite-sample convergence

We compare five finite-sample estimators across sample sizes from 200 to 10,000 trajectories (3 seeds each). Reduced-form logit targets A^*, which is the wrong objective for Type B counterfactuals. IQ-Learn (χ^2) optimizes the Garg et al. (2021) tabular objective with L-BFGS-B. AIRL trains an adversarial discriminator without anchor constraints, so shaping ambiguity persists in finite samples as in Analysis A. LSW-AIRL enforces the two anchor constraints (exit action 2 pinned to zero reward, absorbing state s_{abs} pinned to zero value), achieving full identification at population. GLADIUS uses neural Q and EV networks with 150 training epochs.

N	RF	IQ-Learn (χ^2)	AIRL	LSW	GLADIUS	NFXP
200	0.265	0.237	0.106	0.105	0.132	–
500	0.264	0.287	0.106	0.105	0.135	–
2,000	0.263	0.285	0.106	0.105	0.139	–
5,000	0.262	0.335	0.106	0.105	0.135	–
10,000	0.262	0.333	0.106	0.105	0.130	–

Table 4: Type B CF error ($k = 3$, 3 seeds). AIRL and LSW-AIRL achieve the lowest error (0.106 and 0.105), stable across all sample sizes. GLADIUS is close at 0.13. Tabular IQ-Learn has local optima and degrades at large N. Reduced-form plateaus at 0.262 (wrong target). NFXP pending.

5.5 E. Anchor misspecification

We set the true exit payoff to ε while the anchors assume it is zero. All five methods are evaluated at the population level. The reduced-form and AIRL-no-anchors errors are not affected by the anchor misspecification because they do not use anchors. The error shown is for AIRL with anchors, which degrades smoothly.

ε	Corr. with r^*	Type B CF error	In-sample π shift
0.0	1.000	0.000	0.000
0.5	0.976	0.002	0.000
1.0	0.909	0.004	0.000
2.0	0.729	0.008	0.000

Table 5: Anchor misspecification for AIRL with anchors. Degradation is smooth. In-sample CCPs are unaffected because the policy is identified regardless of reward levels.

5.6 F. Content-based paywall

High-content episodes become paid-only and the wait action is blocked, forcing buy-or-exit. This is a Type B shift that mirrors Section 7.3 of LSW. All five methods are evaluated at the population level.

Method	CCP error
Oracle	0.000
AIRL+anchors	0.000
IQ-Learn / GLADIUS	0.000
AIRL-no-anchors	0.009
Reduced-form	0.185

Table 6: Content-based paywall counterfactual. The reduced-form approach fails at 18.5pp because it cannot separate content utility from continuation value.

5.7 G. State-only reward (Fu et al. transfer test)

Analyses A through F use action-dependent rewards, which is the LSW setting. Here we test the Fu et al. (2018) setting by replacing the true reward with a state-only function $r^*(s, a) = r^*(s)$ for

all a, where $r^*(s) = u(c)$ depends only on content quality. This satisfies the conditions of Fu et al. Theorem 5.1, so AIRL with a state-only $g_\theta(s)$ should recover $r(s)$ up to a constant that cancels in the softmax. We evaluate the Type B counterfactual with $k = 3$, re-solving the Bellman equation without pinning $V(s_{\text{abs}}) = 0$ (the absorbing-state anchor is not part of the Fu et al. identification strategy).

Method	MSE vs. r^*	Corr.	Type B err
Oracle / AIRL+anchors / IQ-Learn	0.000	1.000	0.000
AIRL-state-only ($g^* = r + \text{const}$)	2.778	1.000	0.000
AIRL-shaped ($\alpha = 0.5$)	13.689	0.530	0.026
Reduced-form	54.755	0.000	0.182

Table 7: State-only reward DGP. AIRL with state-only $g_\theta(s)$ recovers $r^*(s)$ up to a constant (MSE > 0 but Corr $= 1$) and transfers perfectly (Type B error $= 0$). This is the mechanism behind the Fu et al. sim-to-real experiments. Generic AIRL without the state-only restriction still fails (0.026) because the shaping residual is action-dependent. The reduced-form advantage has zero correlation with $r^*(s)$ because the true reward is identical across actions and the advantage reflects only continuation-value differences.

6 Summary

	RF	AIRL (no anchors)	AIRL (anchors)	IQ-Learn	GLADIUS	NFXP
Uses $P(s'\|s,a)$	No	No (sim.)	No (sim.)	Yes (exact)	No (empirical)	Yes (exact)
Recovers	A^*	r + shaping	r^*	r^*	r^*	r^*
Type A CF	✓	✓	✓	✓	✓	✓
Type B CF	×	×	✓	✓	✓	✓

Table 8: Summary of six approaches to reward recovery.

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